

Consumption of eggs with meals increases the susceptibility of human plasma and low-density lipoprotein to lipid peroxidation.

Consumption of eggs for a long period was shown to result in hypercholesterolemia and is generally restricted for this reason. In the present study we analyzed the effect of eggs consumption for 3 weeks on lipoprotein atherogenicity. Consumption of 2 eggs per day with the meals, for 3 weeks resulted in a minor elevation in plasma glucose and urea concentrations. Plasma cholesterol concentration increased by 11% ($p < 0.05$) as a result of increased plasma low-density lipoprotein (LDL) cholesterol levels. Plasma triglycerides decreased by 13% ($p < 0.01$), but there were no significant alterations in plasma apolipoproteins A-I or B-100 concentrations. Plasma high-density lipoprotein (HDL) cholesterol decreased by 11% ($p < 0.05$). There was a 13% reduction, though not significant, in the cholesterol efflux from J-774 A.1 macrophages by HDL that was derived after eggs consumption in comparison to HDL that was obtained at baseline. The susceptibility of plasma [using 100 mM of 2,2' azobis 2-amidinopropane (AAPH)] as well as that of LDL (using 10 microM of copper ions) to lipid peroxidation was increased by 42% and 34%, respectively, as measured by the thiobarbituric acid reactive substance (TBARS) assay ($p < 0.01$). Kinetic analysis of LDL oxidation by copper ions revealed a 37% reduction in the lag time required for the initiation of LDL oxidation after 3 weeks of eggs consumption. The total plasma fatty acids concentration increased from 2.2 +/- 0.5 to 3.2 +/- 0.6 mg/ml. The plasma antioxidants, vitamin E and carotenoids were not significantly affected by eggs consumption. We conclude that eggs consumption, in addition to its hypercholesterolemic effect, increases plasma and LDL oxidizability, a phenomenon which was shown to enhance the progression of atherosclerosis. The atherogenic properties may contribute to the accelerated atherosclerosis prevalent in populations with high cholesterol intake.